NAME: PARIDHI BAJPAI

SRN: PES2UG24CS337

SECTION: F

**1. Requirements Table**

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| FR-001 | Functional | The system shall cross-reference passenger lost baggage reports against unrouted luggage scan logs across all connected airport terminals. | High | Pass: Match identified and recovery alert generated.<br>Fail: Incorrect bag tag format accepted. | Given as sample requirement — core matching engine of the platform. |
| FR-002 | Functional | The system shall allow a passenger to submit a lost baggage claim with flight number, bag description, and last known scan location. | High | Pass: Claim is saved and a unique Claim ID is returned within 5 seconds.<br>Fail: Claim submitted without a valid flight number. | Passengers need a formal way to initiate the recovery process. |
| FR-003 | Functional | The system shall display a real-time bag tag scan timeline to the passenger, showing every terminal checkpoint the bag has passed through. | Medium | Pass: Timeline shows all scan events in chronological order with terminal names and timestamps.<br>Fail: Timeline skips a logged scan event. | Reduces support calls by letting passengers self-track. |
| FR-004 | Functional | The Baggage Service Agent shall be able to update the status of a claim (e.g., Under Investigation, Located, Delivered). | High | Pass: Status change is saved and passenger is notified within 1 minute. | Keeps passengers informed and gives agents a workflow to manage cases. |

| Req ID | Type | Description | Priority | Acceptance Criteria | Rationale |
|---|---|---|---|---|---|
| | | | | Fail: Status changed without a valid claim ID. | |
| FR-005 | Functional | The system shall calculate and initiate automated passenger compensation when a claim remains unresolved beyond the airline's SLA window (e.g., 96 hours). | Medium | Pass: Compensation record is generated automatically once the SLA timer expires.<br>Fail: Compensation triggered before the SLA window ends. | Automates a costly manual process and improves passenger trust. |
| NFR-001 | Non-Functional (Performance & Security) | The baggage tracking timeline API must return complete multi-leg scan history in under 200 ms. | High | Pass: Benchmarking tests confirm target latency and security standards under simulated peak load. | Given as sample NFR — ensures the portal feels responsive at scale. |
| NFR-002 | Non-Functional (Availability) | The claim submission service shall maintain 99.9% uptime, including during peak travel periods. | High | Pass: Uptime monitoring logs show ≥99.9% availability over a 30-day rolling window.<br>Fail: Any unplanned downtime exceeding the 0.1% error budget. | Baggage loss is highest during peak travel — the system can't go down exactly when it's needed most. |

**2. UML Use-Case Diagram**

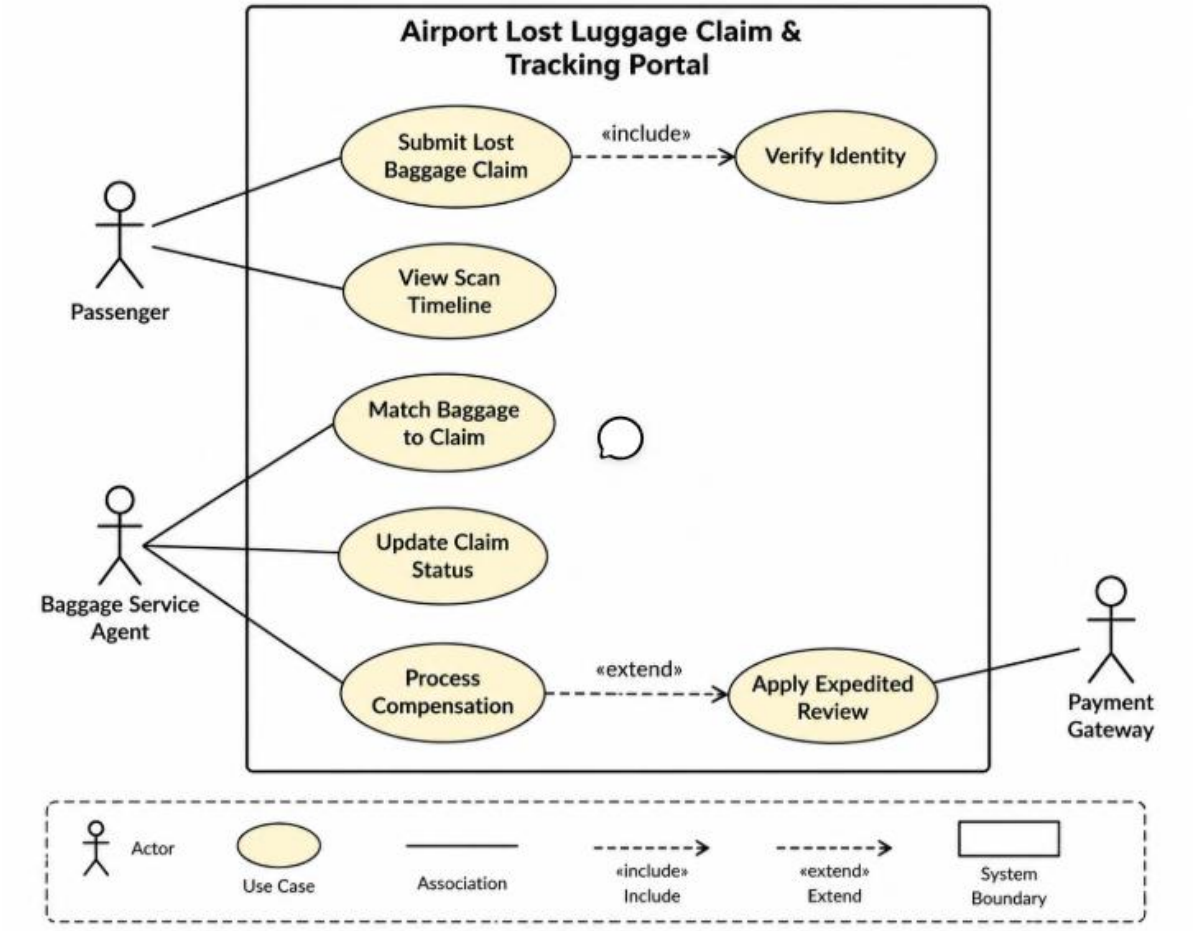
**Actors:** Passenger, Baggage Service Agent, Payment Gateway (external system)

**Use Cases:** Submit Lost Baggage Claim, View Scan Timeline, Match Baggage to Claim, Update Claim Status, Process Compensation, Verify Identity, Apply Expedited Review

**Relationships:**

- Submit Lost Baggage Claim **«include»** Verify Identity (every claim submission always verifies who's filing it)
- Process Compensation **«extend»** Apply Expedited Review (only kicks in optionally, if the claim is flagged urgent)
- Passenger — Submit Lost Baggage Claim, View Scan Timeline
- Baggage Service Agent — Match Baggage to Claim, Update Claim Status, Process Compensation
- Payment Gateway — Process Compensation

---



**3. Use-Case Flow: Submit Lost Baggage Claim**

**Goal:** Passenger successfully files a lost baggage claim and receives a trackable Claim ID.

**Actor:** Passenger

**Preconditions:** Passenger has landed and confirmed their bag did not arrive at baggage claim. Passenger has their boarding pass/flight details available.

**Postconditions:** A claim record exists in the system with a unique Claim ID, status set to "Submitted," and it is now visible to Baggage Service Agents for matching.

**Main Success Scenario:**

1. Passenger selects "Report Lost Baggage" on the portal.
2. System prompts for flight number, date, and passenger identity details.
3. Passenger enters the requested information.
4. System verifies passenger identity against the airline's booking record (include: Verify Identity).
5. Passenger describes the bag (color, brand, tag number if known) and last known checkpoint.
6. System validates the bag tag format and flight details.
7. System generates a unique Claim ID and timestamps the submission.
8. System runs an initial cross-reference against unrouted scan logs (FR-001).
9. Passenger receives on-screen confirmation and an email with the Claim ID and tracking link.
10. Use case ends successfully.

**Alternate Flow — 6a. Invalid Bag Tag Format:**

- 6a1. System displays an error indicating the bag tag number doesn't match a recognized format.
- 6a2. Passenger is prompted to re-enter the tag number or skip it if unavailable.
- 6a3. If the passenger skips it, the claim proceeds with a "tag unknown" flag; use case resumes at step 7.